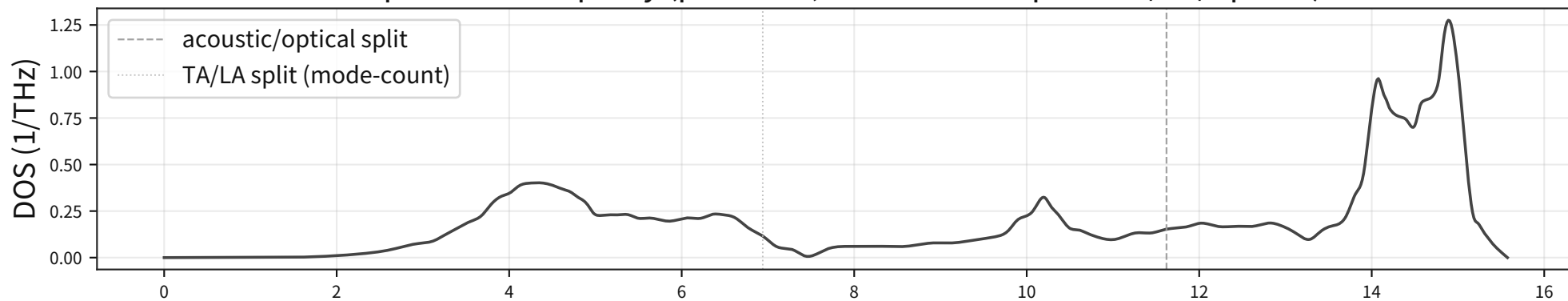


Si phonon DOS proxy (per atom; mode-count splits: TA/LA/optical)



Si thermal expansion: phonon DOS constrained Gruneisen check

